

BODY ELECTRICAL SYSTEM

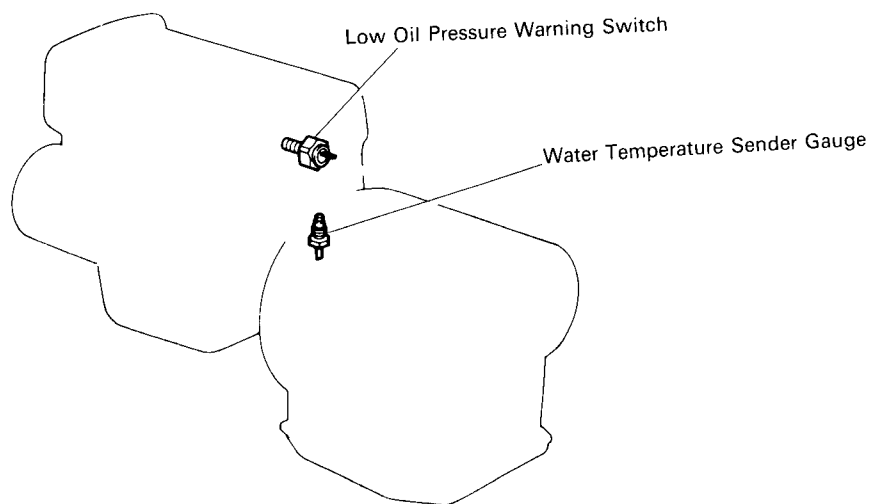
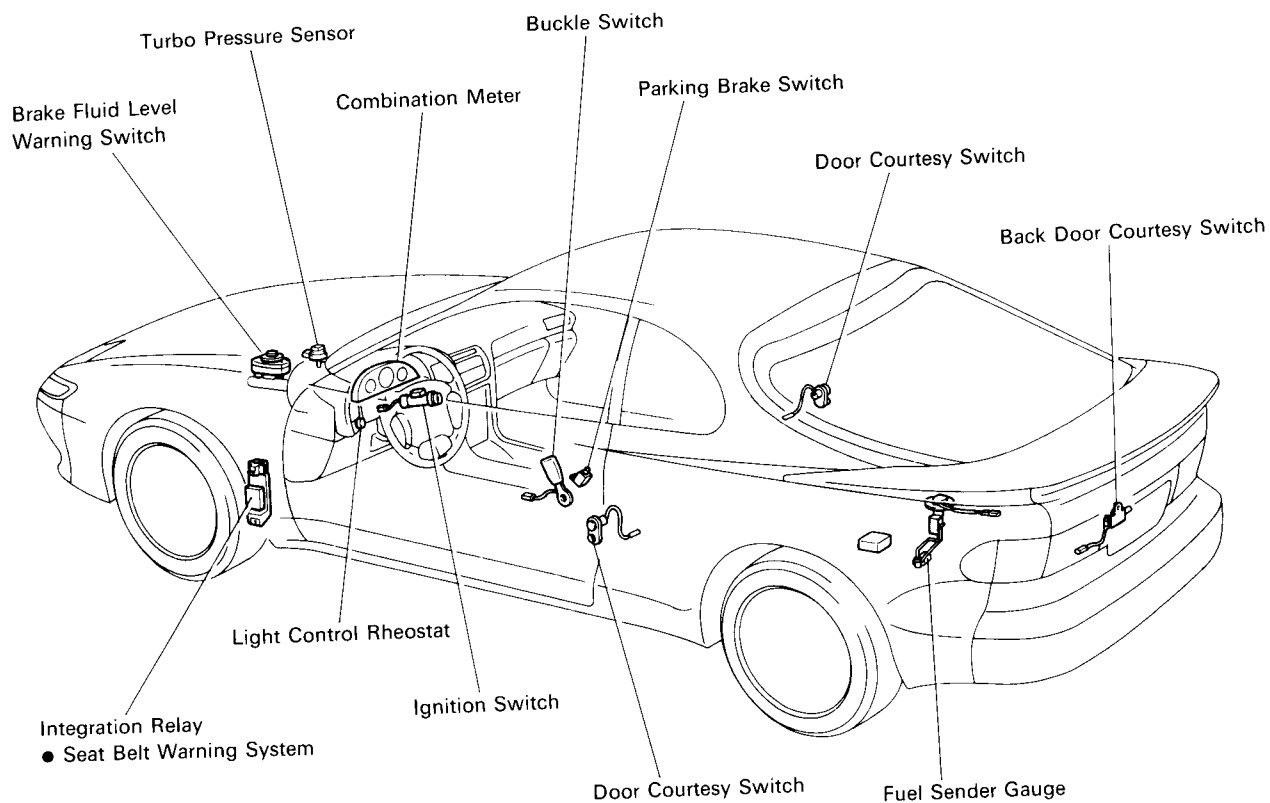
REFER TO 1990 CELICA REPAIR MANUAL (Pub. No. RM149U)

NOTE: The following pages contain only the points which differ from the above listed manual.

	Page
COMBINATION METER	BE-2

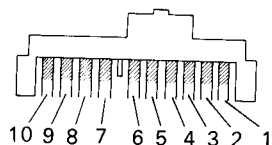
COMBINATION METER

Parts Location

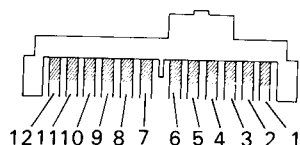


Meter Circuit

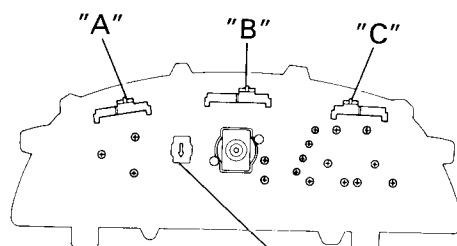
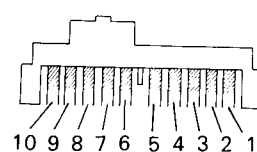
Connector "A"



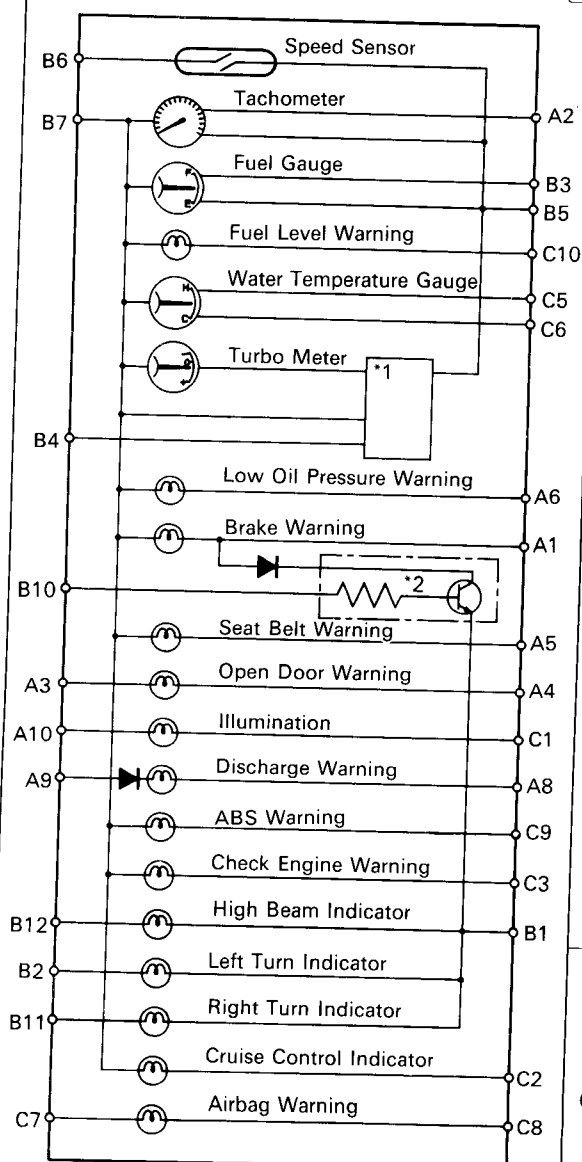
Connector "B"



Connector "C"



Bulb Check Relay



- *1 Turbo Meter Drive Circuit
*2 Bulb Check Relay

BE4015 BE4016 BE4017
BE4689
BE4535

No.	Wiring connector side
A	1 Brake Fluid Level Warning Switch-terminal 1, and Parking Brake Switch
	2 Igniter
	3 Fuse DOME
	4 Door Courtesy Switch
	5 Integration Relay-terminal 4
	6 Low Oil Pressure Warning Switch
	8 Alternator-terminal "L"
	9 Fuse IGN
	10 Fuse TAIL
B	1 Ground
	2 Headlight Dimmer and Turn Signal Switch, and Hazard Warning Switch
	3 Fuel Sender Gauge-terminal 3
	4 Turbo Pressure sensor-terminal 2
	5 Ground
	6 Speed Control Unit
	7 Fuse Gauge
	10 Starter Relay
	11 Headlight Dimmer and Turn Signal Switch, and Hazard Warning Switch
	12 Headlight Dimmer and Turn Signal Switch, and Headlight LO (LH)
C	1 Light Control Rheostat-terminal 3
	2 Cruise Control ECU
	3 Engine ECU
	5 Ground
	6 Water Temperature Sender Gauge
	7 Fuse ECU-B
	8 Airbag ECU
	9 ABS ECU
	10 Fuel Sender Gauge-terminal 2

Troubleshooting

Problem	Possible cause	Remedy	Page
* Gauges and indicator lights do not operate	"GAUGE" fuse faulty Wiring or ground faulty	Replace fuse and check for short Repair as necessary	
Turbo meter does not operate	Turbo meter faulty Turbo pressure sensor faulty Turbo meter drive circuit faulty Wiring or ground faulty	Check turbo meter Check sensor Check drive circuit Repair as necessary	BE-7 BE-7, 8 BE-8
Fuel gauge does not operate	Receiver gauge faulty Sender gauge faulty Wiring or ground faulty	Check gauge Check gauge Repair as necessary	BE-4, 5 BE-5
Fuel level warning light does not light up	Bulb burned out Warning switch faulty Wiring or ground faulty	Replace bulb Check gauge Repair as necessary	BE-5, 6
Water temperature gauge does not operate	Receiver gauge faulty Sender gauge faulty Wiring or ground faulty	Check gauge Check gauge Repair as necessary	BE-6 BE-6

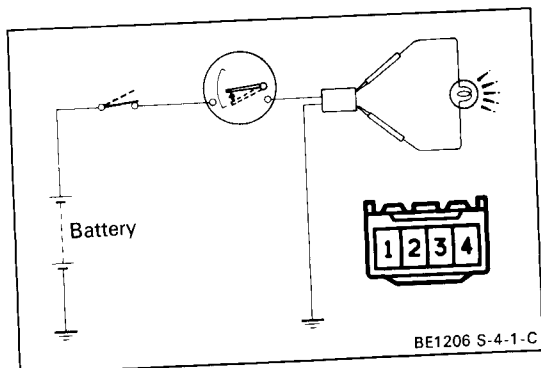
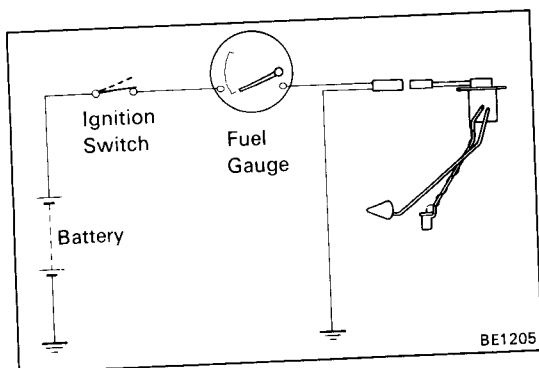
* Tachometer, Fuel Gauge, Fuel Level Warning Light, Water Temperature Gauge, Turbo Meter, Low Oil Pressure Warning Light, Brake Warning Light, Seat Belt Warning Light, ABS Warning Light, Check Engine Warning and Cruise Control Indicator light

Parts Inspection

Fuel Gauge System

1. INSPECT RECEIVER GAUGE (Operation)

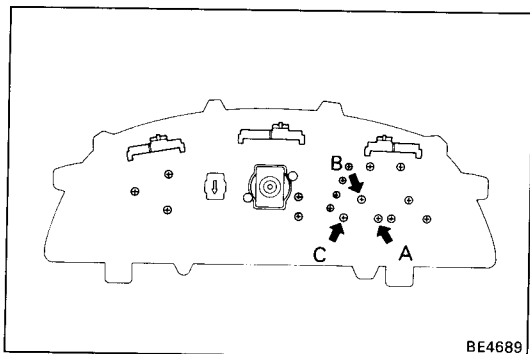
- Disconnect the connector from the sender gauge.
- Turn the ignition switch ON, Check that the receiver gauge needle indicates EMPTY.



- Connect terminals 3 and 4 on the wire harness side connector through a 3.4 watts test bulb.
- Turn the ignition switch ON, check that the bulb lights up and receiver gauge needle moves toward the full side.

HINT: Because of the silicon oil in the gauge, it will take a short time for the needle to stabilize.

If operation is not as specified, inspect the receiver gauge resistance.



BE4689

(Resistance)

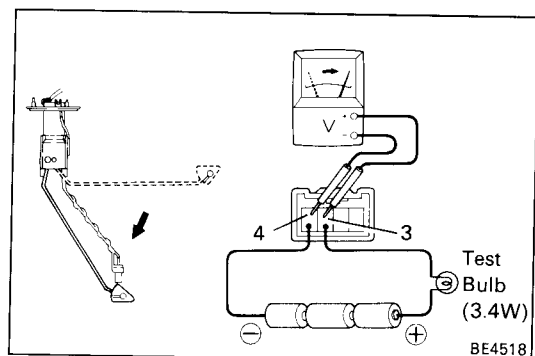
Measure the resistance between terminals.

Between terminals	Resistance (Ω)
A – B	Approx. 83
A – C	Approx. 234
B – C	Approx. 151

If resistance value is not as specified, replace the receiver gauge.

2. INSPECT SENDER GAUGE**(Operation)**

- Connect a series of three 1.5 V dry cell batteries.
- Connect the positive (+) lead from the dry cell batteries to terminal 3 through a 3.4 watts test bulb and the negative (–) lead to terminals 4.
- Check that the voltage rises between terminals 3 and 4 as the float is moved from the top to bottom position.



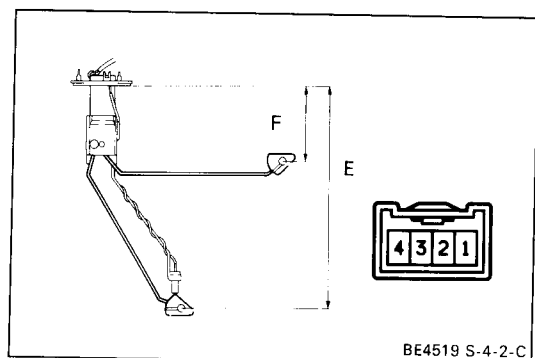
BE4518

(Resistance)

Measure the resistance between terminals 3 and 4 for each float position.

Float position mm (in.)	Resistance (Ω)
F	Approx. 97.1 (3.823)
E	Approx. 272.4 (10.724)

If resistance value is not as specified, replace the sender gauge.

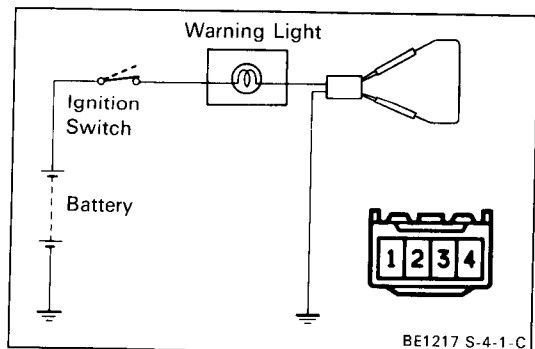


BE4519 S-4-2-C

Fuel Level Warning System**1. INSPECT WARNING LIGHT**

- Disconnect the connector from the sender gauge.
- Connect terminals 2 and 4 on the wire harness side connector.
- Turn the ignition switch ON, check that the warning light lights up.

If the warning light does not light up, test the bulb.

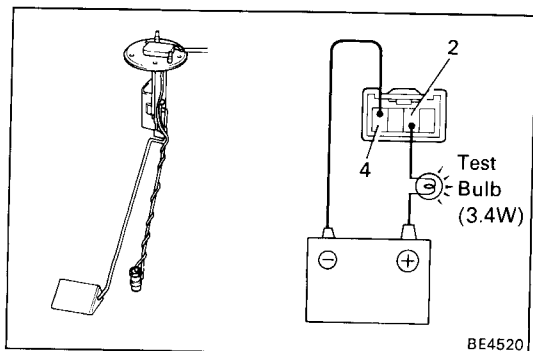


BE1217 S-4-1-C

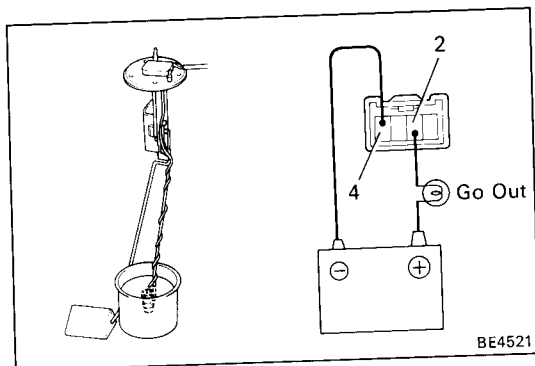
2. INSPECT WARNING SWITCH

- Apply battery voltage between terminals 2 and 4 through a 3.4 watts test bulb, check that the bulb lights up.

HINT: It will take a short time for the bulb to light up.



BE4520



- (b) Submerge the switch in fuel, check that the bulb goes out.

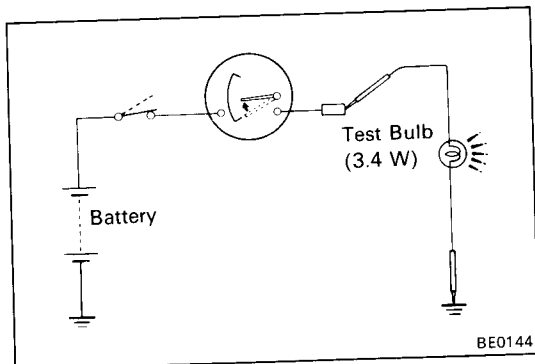
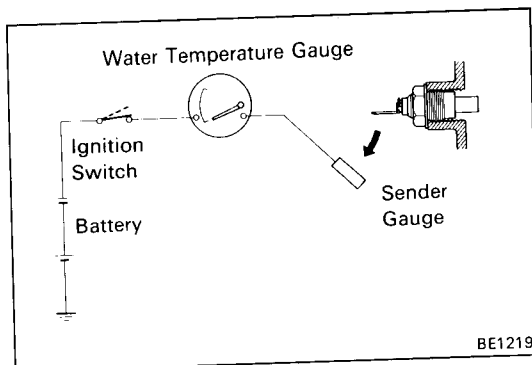
If operation is not as specified, replace the sender gauge.

Water Temperature Gauge System

1. INSPECT RECEIVER GAUGE

(Operation)

- (a) Disconnect the connector from the sender gauge.
 (b) Turn the ignition switch ON, check that the receiver gauge needle indicates COOL.



- (c) Ground terminal on the wire harness side connector through a 3.4 watts test bulb.
 (d) Turn the ignition switch ON, check that the bulb lights up and receiver gauge needle moves toward the hot side.

If operation is not as specified, measure the receiver gauge resistance.

(Resistance)

Measure the resistance between terminals.

HINT: Connect the test leads so that the current from the ohmmeter can flow according to the chart order.

Between terminals	Resistance (Ω)
A $\rightarrow$ B	Approx. 198
A $\rightarrow$ C	Approx. 147
B $\rightarrow$ C	Approx. 51

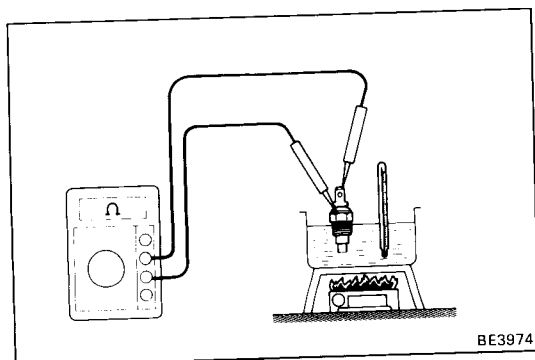
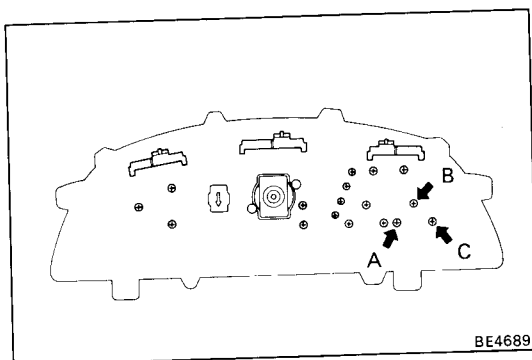
If resistance value is not as specified, replace the receiver gauge.

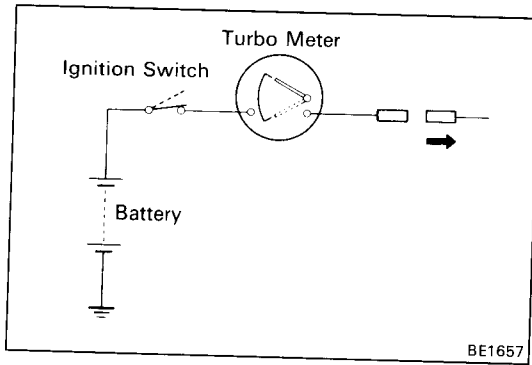
2. INSPECT SENDER GAUGE

Measure the resistance between terminal and the gauge body.

Water temperature $^{\circ}\text{C}$ ($^{\circ}\text{F}$)	Resistance (Ω)
50 (122)	Approx. 226
115 (239)	Approx. 26

If resistance value is not as specified, replace the receiver gauge.





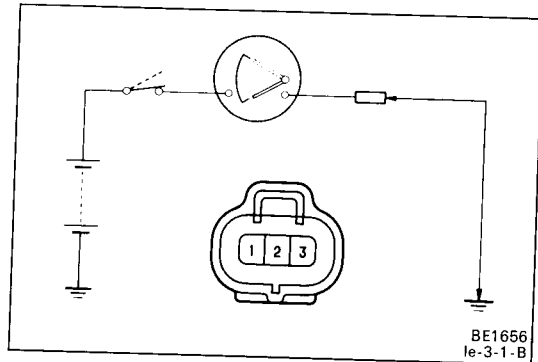
Turbo Meter System

1. INSPECT TURBO METER (Operation)

- Disconnect the connector from the pressure sensor.
- Turn the ignition switch ON. Check that the meter needle moves to upper position.

- Ground terminal 2 on the wire harness side.
Check that the meter needle moves to lower position.

If operation is not as specified, inspect the turbo meter drive circuit and resistance.

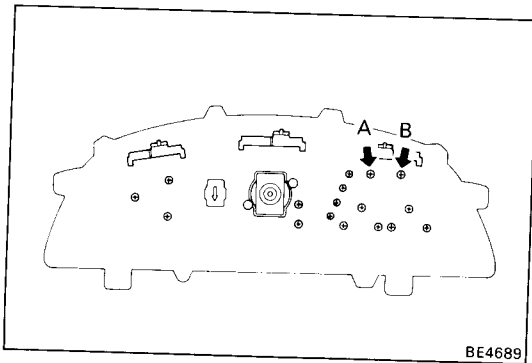


(Resistance)

Measure the resistance between terminals A and B.

Resistance: Approx. 72 Ω

If resistance value is not as specified, replace the turbo meter.



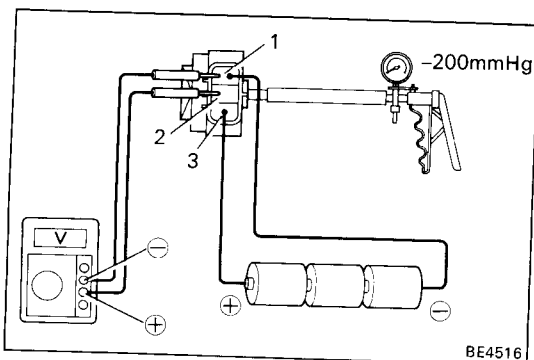
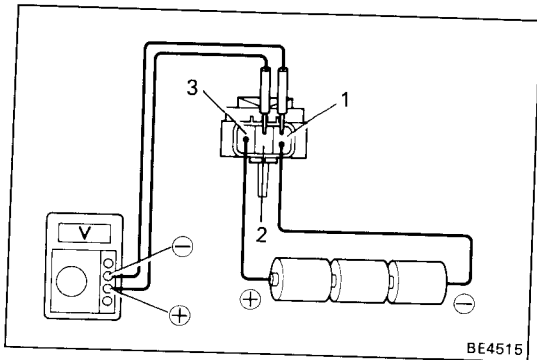
2. INSPECT TURBO PRESSURE SENSOR

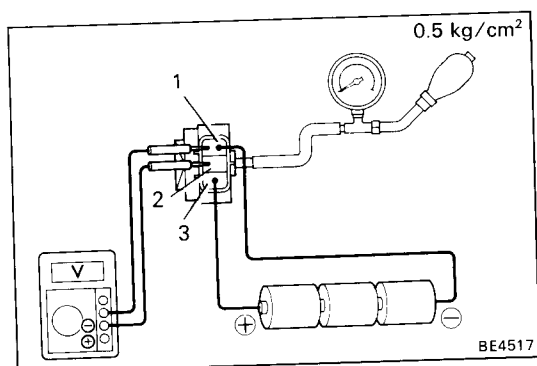
- Connect a series of three 1.5 V dry cell batteries.
- Connect the positive (+) lead from the dry cell batteries to terminal 3 and the negative (-) lead to terminal 1.
- Connect the positive (+) lead from the voltmeter to terminal 2 and the negative (-) lead to terminal 1.
- Check that the voltage between terminals 2 and 1.

Voltage: Approx. 2.4 V

- Apply 200 mmHg (7.87 in.Hg. 26.7 kPa) of vacuum.

Check that the voltage drops below approximately 2.4 V.





- (f) Using SST, apply 0.5 kg/cm² (7.1 psi, 4.9 kPa) of pressure.

Check that the voltage rises approximately 2.4 V.

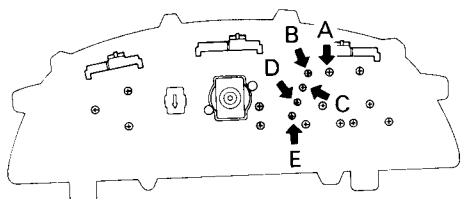
SST 09992-00241

If operations are not as specified, replace the sensor.

**Turbo Pressure Sensor
(Wire Harness Side)**



Combination Meter



1e-3-1-B
BE4689

3. INSPECT TURBO METER DRIVE CIRCUIT

- Disconnect the connector from the turbo pressure sensor.
- Remove the combination meter with connected three connectors.
- Inspect the connector on the wire harness side and terminals of the turbo meter drive circuit as shown in the chart.

Check for	Tester connection	Condition		Specified value
Continuity	A – B	Constant		Continuity
	C – 2	Constant		Continuity
	D – Ground	Constant		Continuity
	I – Ground	Constant		Continuity
Voltage	E – Ground	Ignition Switch position	LOCK or ACC	No voltage
			ON	Battery voltage
	3 – Ground	Ignition switch position	LOCK or ACC	No voltage
			ON	Battery voltage

If circuit is as specified, replace the drive circuit.